

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

Sep 2018

CL 208 Advanced Surveying

Date: 19.09.2018, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Do as directed.

[07]

- 1) What is basic requirement of triangulation station?
- 2) Errors arising from carelessness of the observer are known as
(a) Systematic errors (b) Compensating errors (c) mistakes (d) discrepancy
- 3) The multiplying constant of the tacheometer is given by
(a) f/i (b) $f-i$ (c) $f+d$ (d) $f+2d$
- 4) GPS means
(a) Green Position System (b) Global position Satellites (c) Geographical Public System (d) Global Positioning System
- 5) What is refraction?
- 6) Systematic errors are also known as
a) positive error b) cumulative error c) negative error d) random error
- 7) What is measured by substance bar?

Q - 2 (a) Write any Two.

[14]

- 1) Differentiate between
 - (i) Luminous and non- luminous signals
 - (ii) Systematic errors and accidental errors
- 2) Enlist characteristics of tacheometer. Explain principle of tacheometry.
- 3) Explain the procedure for finding the coefficient in the field for stadia constants K

and C by various methods.

- 4) Define weight and give with examples any four laws of weight.

Q - 3 (a) Write any Two.

[14]

- 1) Why three readings are taken in techeometry? Derive the distance and elevation formulae used in techeometry for staff held vertical and line of sight horizontal.
- 2) **Derive the formulae for a level of a high object**, when Base of the object is inaccessible, Instrument Stations are in the same vertical plane as the elevated object.
- 3) Find the reduced level of the top of church from the given data.

Instrument stations	Reading on B.M	Angle of Elevation	Remark
P	1.578 m	$10^{\circ} 12'$	R.L of B.M= 543.075 m
Q	1.269 m	$8^{\circ} 20'$	

Stations P and Q and the chimney are in the same vertical plane. Take distance between P and Q is 30 m.

- 4) Describe the expression for horizontal and vertical distances by the fixed hair method.

SECTION - II

Q - 4 Define

[07]

Tilt, Isocentre, Side lap, Horizon, Principle Point, Oblique photo, exposure station.

Q - 5 Write any Two.

[14]

- 1) What are the factors that affect the selection of triangulation stations? What considerations you would have while selecting the site for the base line?
- 2) What are the uses of aerial Photogrammetry? Difference between terrestrial and aerial photogrammetry
- 3) What is relief displacement? Derive the equation for it.
- 4) Define with sketch.

(i) hour angle (ii) Nadir (iii) Zenith (iv) Celestial sphere (v) Azimuth (vi) Longitude (vii) Drift

Q - 6 Write any Two.

[14]

- 1) What is GPS? Explain the working principle of GPS
- 2) Discuss all components of GIS and explain the interaction between GIS and GPS
- 3) How remote sensing is useful in civil engineering? Give the types of remote sensing.
- 4) Explain main purposes of field astronomy .Enlist the methods of determination of latitude and explain any one in detail.
